

# Yan Li

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## Research Interests

Trustworthy Medical AI; Multimodal AI for Healthcare; Medical Vision-Language Models; Clinical AI Agents.

## Education

### Beijing University of Posts and Telecommunications

Sept. 2025–present | Beijing, China

*M.S. in Management Science and Engineering*

*Advisor: Yawen Li; Co-advisor: Guanhua Ye.*

### University of Electronic Science and Technology of China

Sept. 2021–July 2025 | Chengdu, China

*Dual B.S. in Finance and Electronic Information Engineering*

*Thesis Advisor: Bo Liu.*

## Research Experience

### CAPRA: Multimodal Auditing under Missing Clinical Metadata

ACM MM 2026 Oral

*Lead Student Researcher, Beijing University of Posts and Telecommunications*

- Led CAPRA, a semantic-visual multimodal framework that fuses medical images with proxy-axis tokens and calibrated posteriors for missing-metadata subgroup auditing.
- Validated fundus, dermoscopy, and chest-radiography tasks under distribution shift, improving worst-group accuracy in 14 of 15 learner-dataset settings.

### Selective-Dependence Auditing for Medical LLM Fairness

Under Review

*Lead Student Researcher, Beijing University of Posts and Telecommunications*

- Led the Selective-Dependence Audit, separating medical judgments, care pathways, support actions, and interface effects to distinguish prohibited dependence from appropriate clinical adaptation.
- Built ClinSDA-400 and evaluated six LLMs across 1,075,020 responses in semantic-label, JSON, and MCQ formats, isolating audit displacement caused by option order and answer-symbol mapping.

### MedFuse: Anatomical-Prior Fusion for Retinal Lesion Segmentation

FCS 2026

*Lead Student Researcher, Beijing University of Posts and Telecommunications*

- Led technical development with an ophthalmologist at Peking University First Hospital, integrating RGB fundus features with retinal vascular priors for diabetic-retinopathy lesion segmentation.
- Evaluated a hierarchical dual-encoder fusion pipeline on DDR and IDRiD across five backbones, including annotation-free priors from a multimodal large language model.

## Peer-Reviewed Publications

### **ACM MM 2026 ORAL** Beyond Metadata: CAPRA for Hidden Subgroup Analysis under Missing Metadata in Medical Imaging

Yawen Li, **Yan Li**, Zhe Xue, Yingxia Shao, Meiyu Liang, and Guanhua Ye<sup>\*</sup>.

*Proceedings of the 34th ACM International Conference on Multimedia, 2026.*

### **ACM MM 2026** What Does Your Short-Answer VQA Score Actually Measure? Evaluator-Dependent Instability in Multimodal Short-Answer Benchmarks

Guanhua Ye, Jingbin Niu, **Yan Li**, Meiyu Liang, Zhe Xue, Yingxia Shao, and Yawen Li<sup>\*</sup>.

*Proceedings of the 34th ACM International Conference on Multimedia, 2026.*

### **AAAI 2026** Rethink Representation Learning for Questionnaire Data

Guanhua Ye, Jifeng He, **Yan Li**, Junping Du, Zhe Xue, Yingxia Shao, Meiyu Liang, and Yawen Li<sup>\*</sup>.

*Proceedings of the AAAI Conference on Artificial Intelligence, 40(33):27782–27790, 2026.*

**TKDE 2025** **FeBT: A Feature Balancing Transformer for Corporate ESG Forecasting**  
Yawen Li, Mengyu Zhuang, Guanhua Ye\*, **Yan Li**, Junheng Wang, Jinyi Zhou, and Pengfei Zhang.  
*IEEE Transactions on Knowledge and Data Engineering*, 2025.

**FCS 2026** **MedFuse: A Multi-source Data Fusion Framework for Diabetic Retinopathy Lesion Segmentation**  
Jiantong Du, **Yan Li**, Yawen Li, Liwen Liao, Zhihao Zhao, and Guanhua Ye\*.  
*Frontiers of Computer Science*, 20(3):2003625, 2026.

Conference Presentations

|                                                                       |                                                |
|-----------------------------------------------------------------------|------------------------------------------------|
| “Trust-free Personalized Decentralized Learning”                      | Feb. 5, 2026                                   |
| Oral Presentation at IEEE BigComp 2026, Guangzhou, China.             | <a href="#">Paper</a>   <a href="#">Slides</a> |
| “Slot-Based Evidence Routing for Budgeted Multi-Agent Coordination”   | May 15–17, 2026                                |
| Oral Presentation at CINT 2026, Hangzhou, China.                      | <a href="#">Paper</a>   <a href="#">Slides</a> |
| “Auditing Historical Clinical Queries with Missing Availability Time” | Aug. 21, 2026                                  |
| Oral Presentation at CCF BigData 2026, Harbin, China.                 | <a href="#">Paper</a>                          |

Teaching

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|-------------------------------------------------------------------------|-------------|
| <b>Beijing University of Posts and Telecommunications</b>               |             |
| <b>Big Data and Intelligent Management</b>                              | Fall 2025   |
| Teaching Assistant, graduate course, School of Economics and Management |             |
| <b>Operating Systems</b>                                                | Spring 2026 |
| Teaching Assistant, undergraduate course, School of Computer Science    |             |

Honors and Awards

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|------------------------------------------------------------------------------|------|
| <b>University of Electronic Science and Technology of China</b>              |      |
| Certificate for Honors Research                                              | 2025 |
| “Academic Star” Award of School of Management and Economics (sole recipient) | 2024 |
| Outstanding Student Scholarship                                              | 2024 |
| Model Student Scholarship                                                    | 2023 |